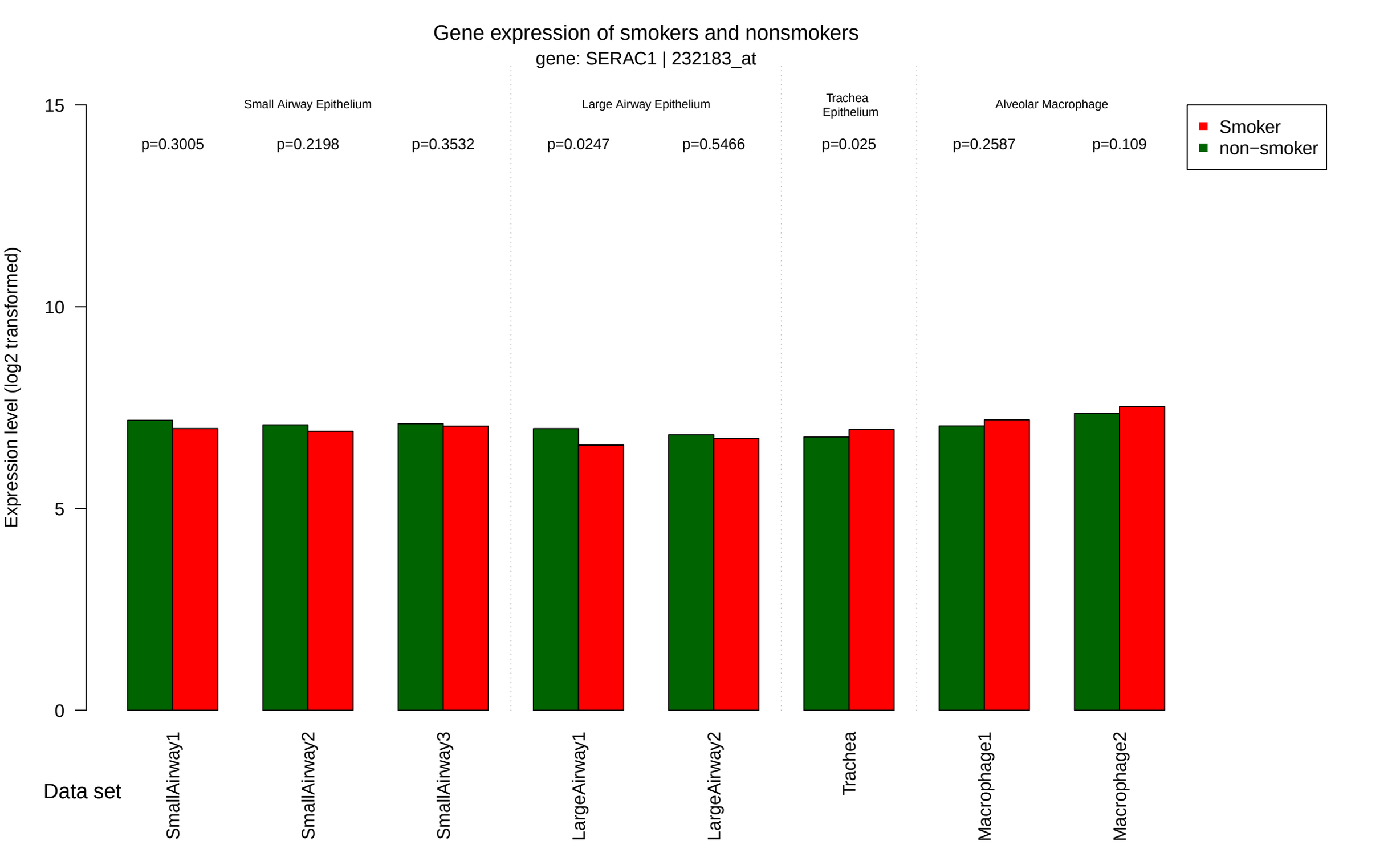
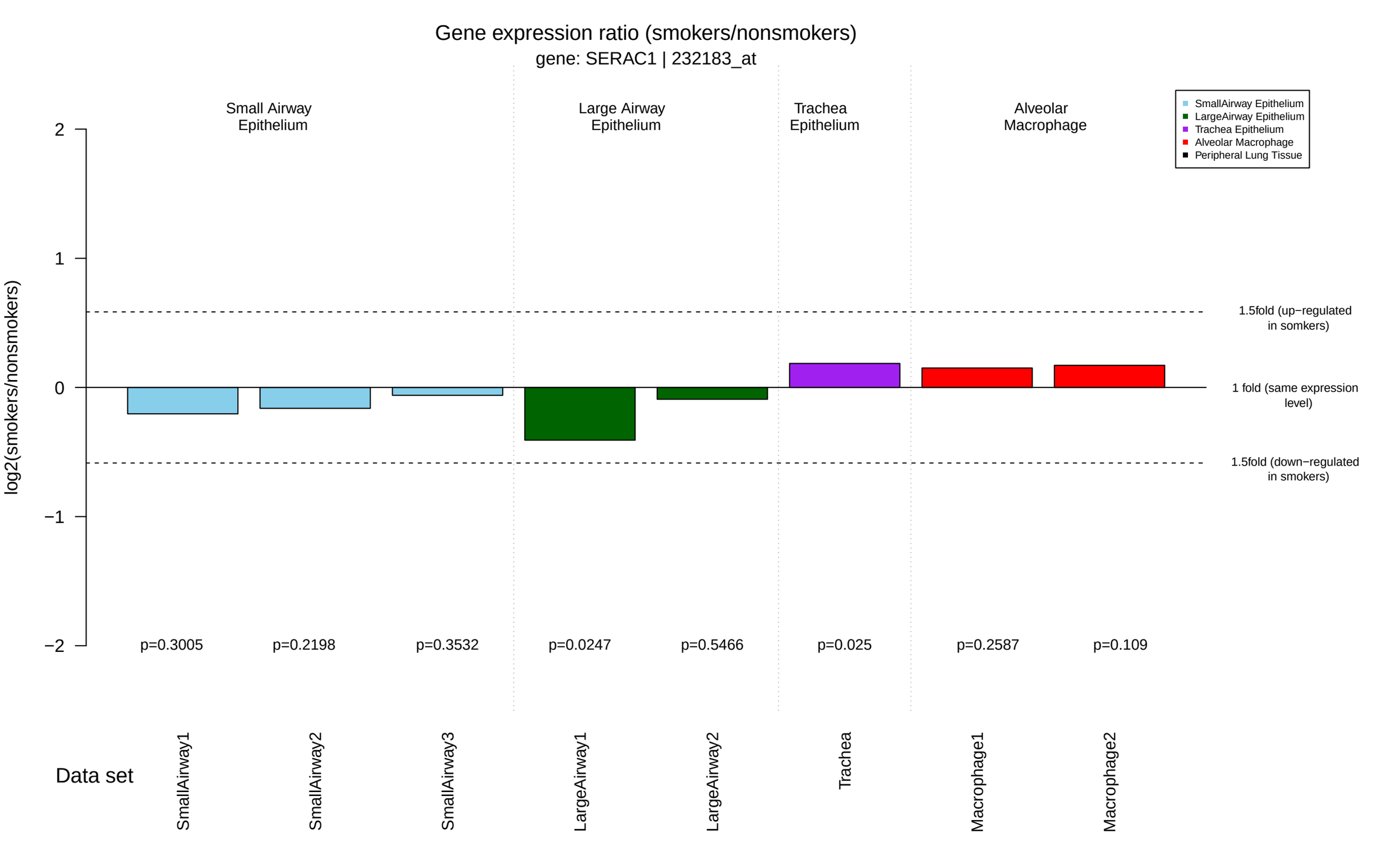
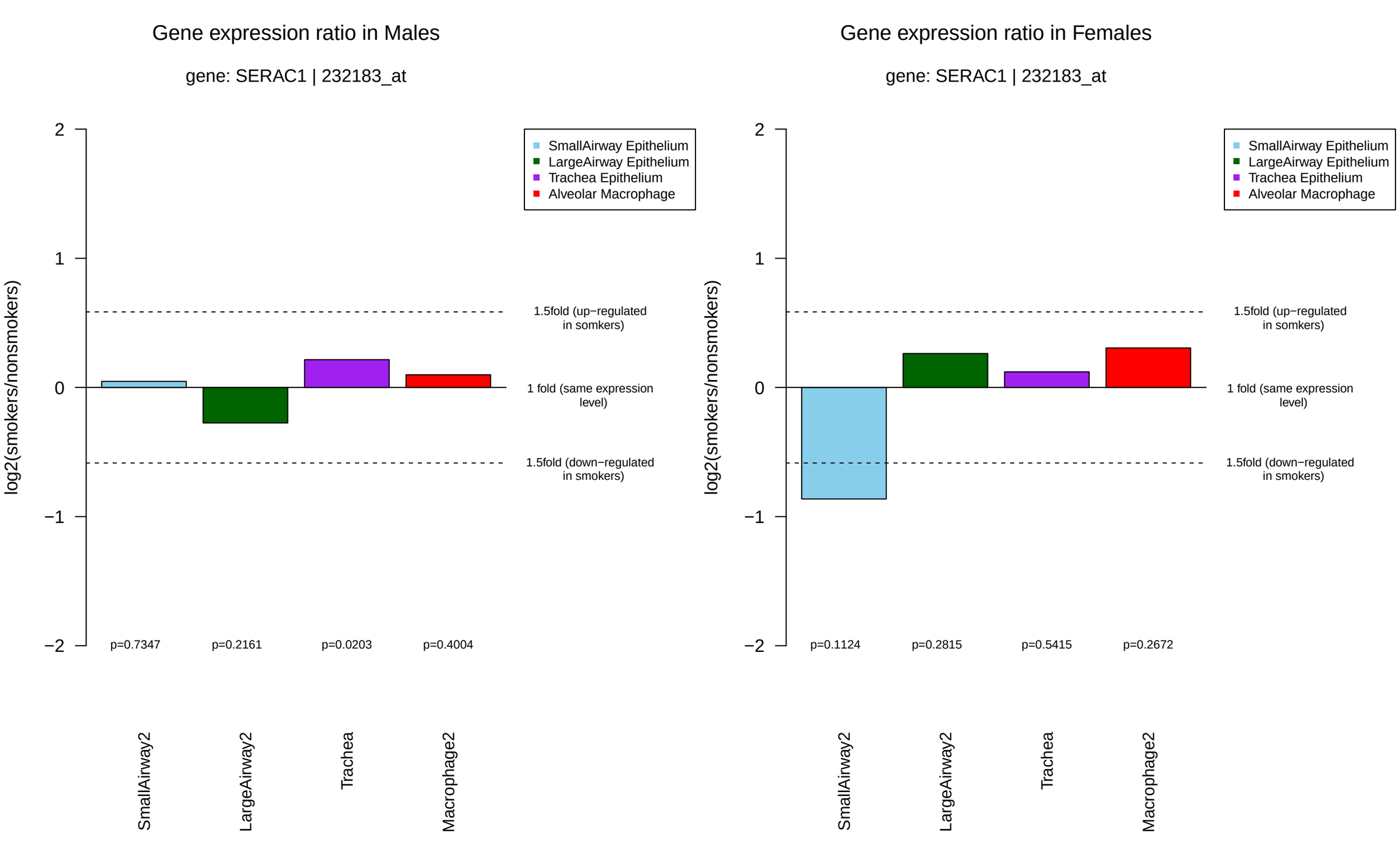
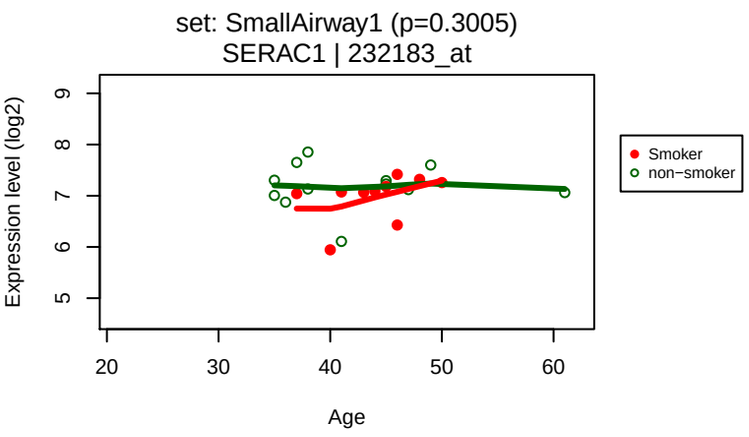
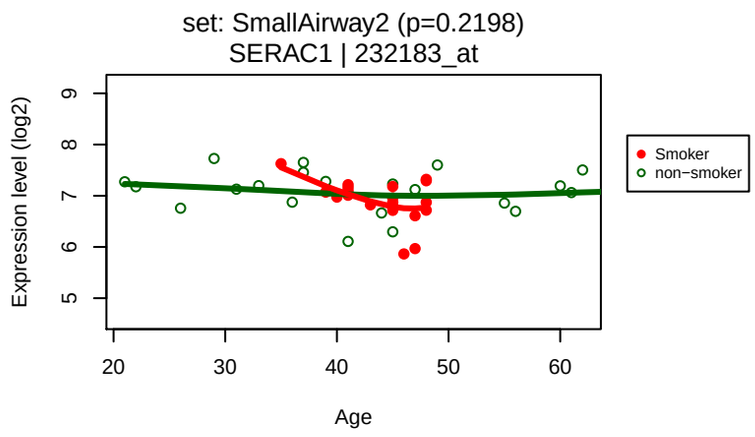
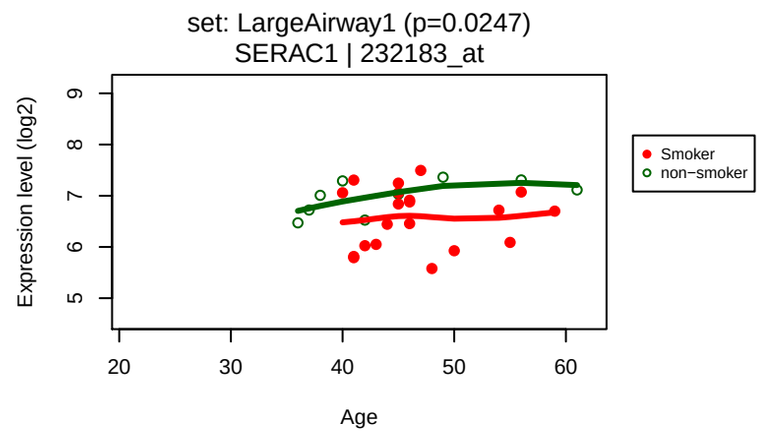

Gene expression of smokers and nonsmokers
gene: SERAC1 | 232183_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3005
p=0.2198
p=0.3532
p=0.0247
p=0.5466
p=0.025
p=0.2587
p=0.109
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

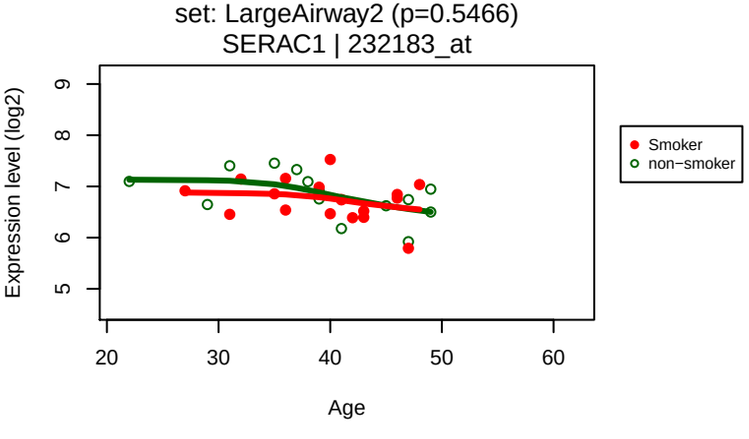

Gene expression ratio (smokers/nonsmokers)
gene: SERAC1 | 232183_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3005
p=0.2198
p=0.3532
p=0.0247
p=0.5466
p=0.025
p=0.2587
p=0.109
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

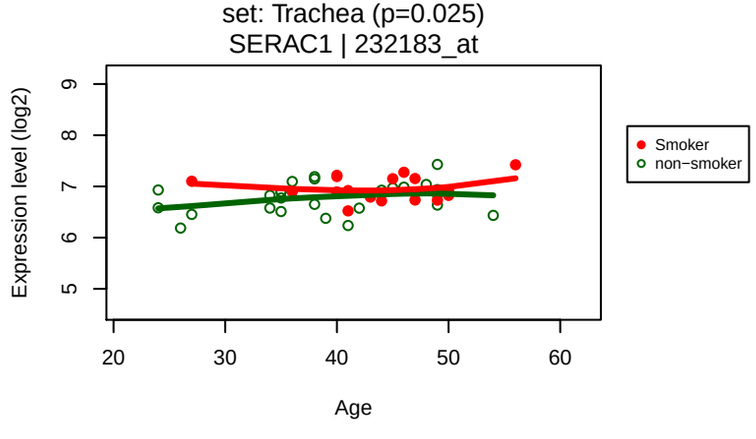

Gene expression ratio in Males
gene: SERAC1 | 232183_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7347
p=0.2161
p=0.0203
p=0.4004
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERAC1 | 232183_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1124
p=0.2815
p=0.5415
p=0.2672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

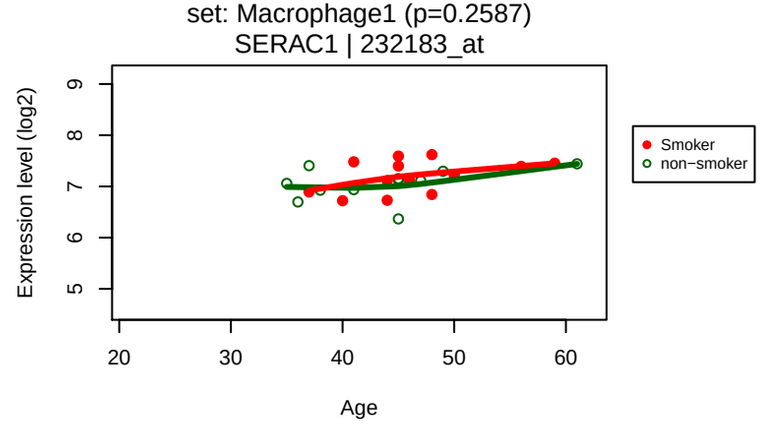

set: SmallAirway1 (p=0.3005)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker

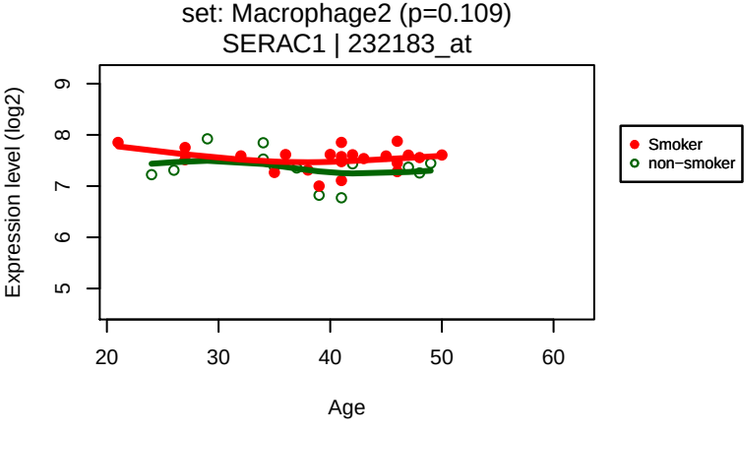

set: SmallAirway2 (p=0.2198)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker

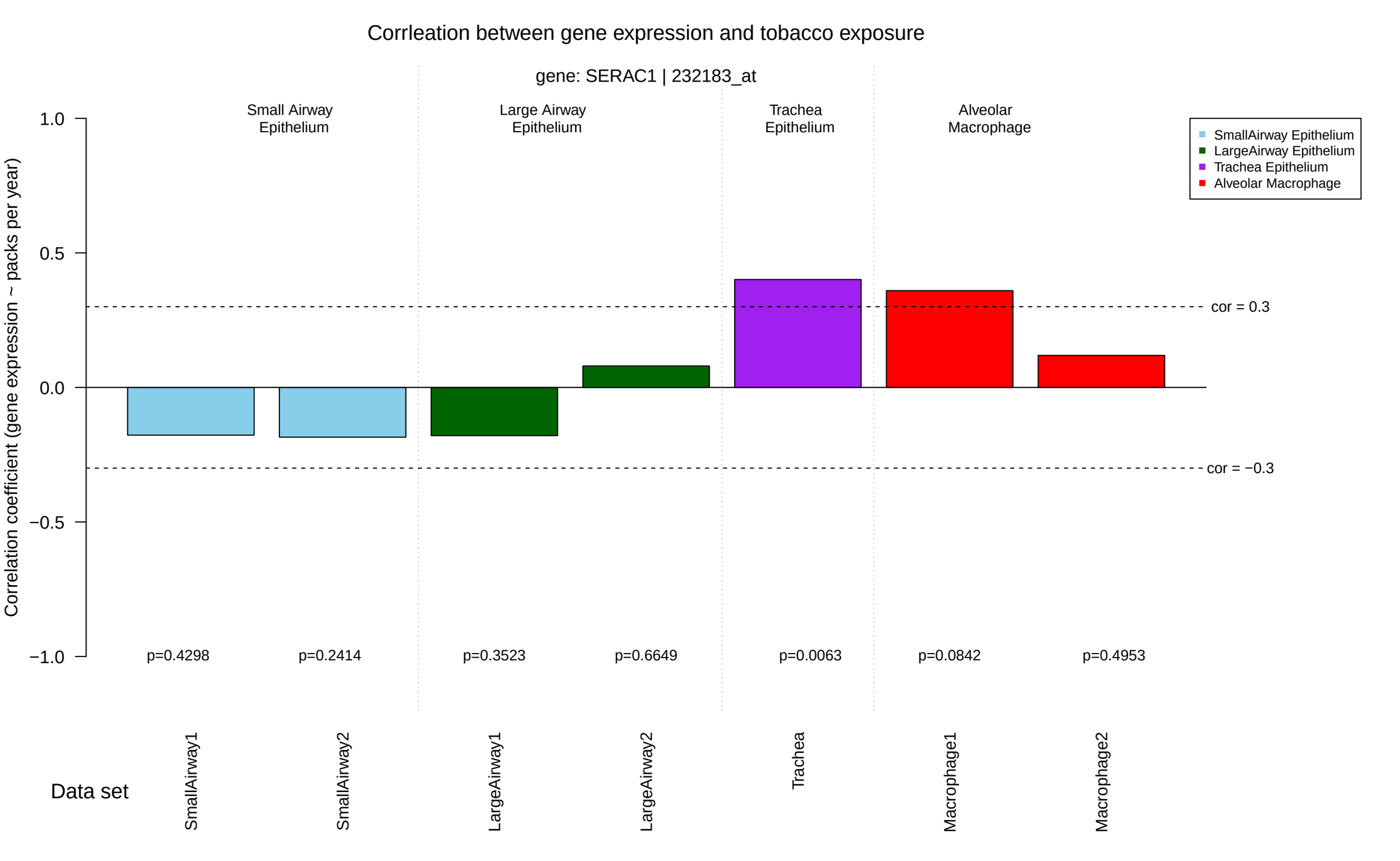

set: LargeAirway1 (p=0.0247)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5466)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.025)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2587)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.109)
SERAC1 | 232183_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERAC1 | 232183_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4298
p=0.2414
p=0.3523
p=0.6649
p=0.0063
p=0.0842
p=0.4953
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2